

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MRBENCH | Context: Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

Input form is a LOWER-priority dimension. MRBench is text-based English dialogue [Q13, Q133], encoded as conversation history with tutor/student turns and lengths measured in characters [Q147]. The deployment is also text-only English-medium via mobile or enterprise software and explicitly targets fluent English speakers; no modality, script, or language mismatch exists. Mobile internet penetration in metropolitan India is high (75%+ of children aged 5–14 use mobile phones, with metro rates higher) [WEB-10, WEB-12], and Android dominates with 92.4% market share [WEB-13], so text-based delivery on mobile is well-supported. The minor concerns are that conversation lengths and turn structures differ between Bridge and MathDial [Q141, Q142] — but this represents internal diversity rather than a target-context mismatch.

ID	Evidence
WEB-10	Mobile internet broadly available in Indian metros for text delivery
WEB-12	~65% of Indian children aged 5–14 used mobile phones in 2023
WEB-13	Android dominates Indian smartphone OS market at 92.4% in 2025

Strengths

- Text-only English format perfectly matches the deployment's text-based English-medium delivery [Q13]
- Conversation-history input format directly matches a mobile-app tutoring dialogue use case [Q13, Q133]
- Internal diversity in conversation length across Bridge and MathDial provides some signal-distribution variation [Q141, Q142]

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are aligned: both benchmark and deployment are English-medium text dialogues with conversation history and turn structure [Q13, Q133]. No script or language mismatch.

IF-2: Regional infrastructure supports the format. India has approximately 750M+ mobile internet users with strong metropolitan 5G coverage [WEB-10, WEB-11], and Android dominates at 92.4% market share [WEB-13].

ID	Evidence
WEB-10	Mobile internet broadly available in Indian metros for text delivery
WEB-11	Mobile broadband dominates Indian internet access; 5G rollout in metros since 2022
WEB-13	Android dominates Indian smartphone OS market at 92.4% in 2025

IF-3: Domain-specific form is well-matched: tutor/student dialogue turns over text are the deployment's actual use case. No multimodal (audio, image, handwriting) requirement is specified.

IF-4: Minor gap: MathDial does not provide conversation topic labels [Q145], which could complicate routing/curriculum-tagging in deployment. Otherwise no form mismatch identified.

Information Gaps

- No data on actual deployment device distribution for Grade 1–8 metropolitan students specifically (smartphone vs. tablet vs. laptop share)

Requires Expert Verification

- Whether deployment intends to support voice/audio modalities later, which would require benchmark extension